

A formal knowledge base for metagenomics and geochemistry of the Rub' al Khali desert

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Abstract

Integrating multi-omic data with environmental metadata remains a significant challenge due to the heterogeneity of data formats and a lack of standardized semantic representations. Building such an integrated resource requires addressing several interlocking challenges: defining a unified data model across instruments and observation types, harmonizing taxonomic identifiers across reference databases, validating incoming data against shared schemas, and supporting semantic query answering at scale under a tractable reasoning regime. Here, we present a high-resolution microbiome dataset comprising soil samples from 60 sampling sites across the Empty Quarter (Rub' al Khali) of Saudi Arabia. We augment these biological sequences with a formal knowledge base constructed using OWL 2 DL and the SemanticScience Integrated Ontology (SIO), with ShEx-based ingest validation and ELK-based materialization over the resulting graph. This semantic layer enables complex cross-domain queries and provides a comparative baseline for studying microbial adaptation in extreme environments.

Keywords: Knowledge Graph, Metagenomics, Rub' al Khali, SIO, Semantic Web

1 Background and summary

The Rub' al-Khali (Empty Quarter) is the largest contiguous sand desert on Earth, and it covers approximately 650,000 km² across the southern Arabian Peninsula [1]. Despite extreme aridity and surface temperatures that exceed 50°C, this environment harbors microbial communities whose composition, diversity, and ecological drivers remain poorly characterized [2]. To predict ecosystem responses to climate change and to inform conservation and land-use policy, researchers require deep knowledge of the biogeochemistry of such extreme environments [3].

To characterize complex ecosystems, researchers must integrate heterogeneous and multi-scale data: microbial community profiles derived from DNA sequencing, elemental soil composition from geochemical analyses, spatiotemporal environmental variables, and provenance metadata that link physical samples to their digital representations. Traditional approaches store these data types in separate, disconnected repositories, which limits cross-domain queries and reduces reproducibility.

Engineers originally developed the concept of a digital twin, and environmental scientists increasingly adopt this approach [4, 5]. Recent frameworks, such as TwinEco, standardize the development of dynamic, data-driven digital twins in ecology to avoid fragmented software designs [6]. Additionally, regional natural resource management benefits from artificial intelligence-assisted digital twin architectures that unify heterogeneous environmental observations [7]. Such digital twins show strong predictive accuracy when they incorporate continuous data streams, such as citizen science observations for real-time biodiversity forecasts [8]. To guarantee that these models remain interoperable and reusable across domains, researchers emphasize the integration of datasets, models, and computational workflows through Findable, Accessible, Interoperable, and Reusable (FAIR) principles [9]. However, while biological systems and specific regions have digital representations [10], researchers have not yet developed semantic representations of entire desert ecosystems. Furthermore, ecosystem digital twins largely focus on macroscopic scale and do not go across levels of granularity.

Constructing such a resource for an entire desert ecosystem requires addressing several interlocking challenges that have shaped the design of this work. First, the data are produced by heterogeneous instruments and protocols (16S amplicon sequencing, X-ray fluorescence, GPS-tagged field notes, gridded climate reanalysis), so a *unified data model* is needed that can represent samples, processes, qualities, and quantities under a common upper-level vocabulary rather than as disconnected tables. Second, taxonomic labels are reported by multiple primer choices and classifier workflows and must be *harmonized* against a single reference taxonomy without losing the original assertions. Third, because data are deposited in batches over five field expeditions, *schema validation at ingest* is required to catch structural errors before they propagate into the graph. Fourth, the knowledge graph must support *semantic query answering at scale*: with tens of millions of triples, expressive OWL 2 DL reasoning is computationally infeasible, so a tractable reasoning regime must be selected and its trade-offs made explicit. Fifth, a *user-facing query and exploration interface* is required so that domain scientists who are not fluent in SPARQL can interact with the resource. We return to each of these challenges in the usage notes (Section 6) and link them to the corresponding design choices in the methods and knowledge representation sections.

Here, we develop a knowledge graph of large parts of the Rub’ al-Khali desert ecosystem to serve as the semantic foundation of the Rub’ al-Khali digital twin. We present an OWL 2 DL knowledge graph [11] that contains 37.4 million RDF triples [12]. This knowledge graph integrates data from five field expeditions (2023–2025) across 70 sampling sites that span an over 1,000 km transect from the western to the east of the Rub’ al-Khali. The dataset includes 16S rRNA gene amplicon profiles from 2,535 soil samples, which yield 1,401,008 taxon-sample abundance observations. We incorporated X-ray fluorescence (XRF) measurements of 91 chemical analytes

from 249 measurement sessions. Furthermore, we integrated site-specific environmental metadata recorded at each visit and monthly climate records derived from the Open-Meteo database [13].

We structured the knowledge graph with the SemanticScience Integrated Ontology (SIO) [14] as the upper-level framework. This design allows interoperability with other SIO-based datasets. We modeled domain-specific concepts as subclasses of SIO entities, and all measurements follow SIO’s ontology design pattern for measurement processes. We grounded chemical entities in the Chemical Entities of Biological Interest (ChEBI) ontology [15], environmental contexts in the Environment Ontology (ENVO) [16], and taxonomic identifiers in the NCBI Taxonomy [17].

Because it encodes the full provenance chain from physical sample collection to digital data products, the knowledge graph is the key foundation for implementing the FAIR data principles for the Rub’ al-Khali digital twin. This semantic integration eliminates the data fragmentation that typically limits ecological models, and it guarantees that observations, models, and workflows remain fully interoperable [9]. Consequently, the knowledge graph acts as the operational foundation for the Rub’ al-Khali digital twin. It allows researchers to execute cross-domain queries that otherwise demand complex joins across multiple heterogeneous tables in a conventional database. Therefore, this resource serves as a foundational infrastructure for computational ecology, which facilitates hypothesis generation, intervention simulation, and evidence-based policy.

This resource makes several contributions. On the scientific side, it provides a coordinated multi-trip desert microbiome and geochemistry dataset—16S rRNA amplicon profiles, XRF elemental measurements, GPS-tagged field metadata, and monthly climate records—for 2,535 soil samples from 70 sites across a 1,000 km transect of the Rub’ al-Khali, together with a harmonized taxonomic abundance table that maps operational taxonomic units from heterogeneous classifier outputs onto NCBI Taxonomy while preserving the original assertions. On the technical side, the resource comprises an OWL 2 DL knowledge graph (37 million asserted triples, expanded to approximately 118 million triples after OWL-Horst materialization) built on SIO with domain classes grounded in ChEBI, ENVO, and NCBI Taxonomy, modelling the full provenance chain from field sampling to digital data products. We formalize reusable ontology design patterns for sampling, measurement, taxonomic abundance, and provenance aligned with SIO’s core idioms. A ShEx-based ingest validation workflow enforces structural shapes on every RDF batch before loading, and the ELK reasoner is used to validate logical consistency of the knowledge graph. OWL-Horst forward-chaining materialization in Virtuoso expands subclass, subproperty, inverse, and transitivity entailments so that query answering can be served from physically materialized triples. A set of competency questions documents the intended query workload and serves as regression tests, and a web portal with an interactive map, per-site dashboards, and a SPARQL editor exposes the knowledge graph to domain scientists without requiring SPARQL fluency.

2 Methods

2.1 Study area and sampling design

We conducted five sampling expeditions across the Rub' al-Khali along a transect from the western escarpment (19.15°N, 45.17°E) to the eastern border of Saudi Arabia with Oman (20.71°N, 54.73°E), spanning 1,043 km. Figure 1 illustrates the altitude profile across this transect.

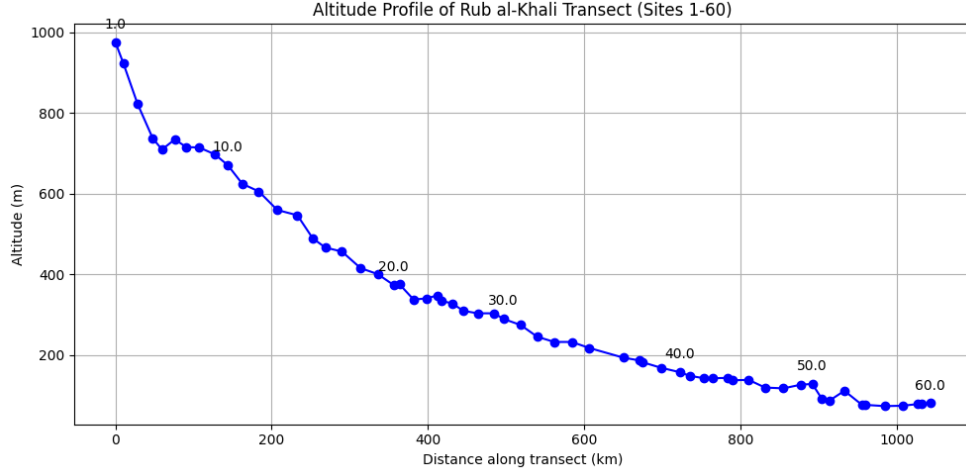


Fig. 1 Altitude profile along the sampling transect across the Rub' al-Khali, from the western escarpment to the eastern border with Oman.

The transect encompasses 60 primary sampling sites at predetermined geographic coordinates, alongside 10 sites of special interest (water wells, a hot spring, an oil spill site), totaling 70 sampling sites.

To capture temporal variability, we conducted expeditions across different seasons: late winter 2023 (Trip 1, 64 sites visited), summer 2023 (Trip 2, truncated to 8 sites due to extreme temperatures), winter 2024 (Trip 3, 65 sites visited), summer 2024 (Trip 4, 60 sites visited), and autumn 2025 (Trip 5, 62 sites visited). We summarize the observational coverage in Table 1. We recorded temperature, atmospheric pressure, and relative humidity at each visited site. Furthermore, we gathered X-ray fluorescence (XRF) measurements during one expedition to determine geochemical soil properties.

At each site, we collected samples from three distinct soil compartments: bulk surface soil (0 cm depth), bulk subsurface soil (5–10 cm depth, hereafter “deep”), and subsurface soil near plant roots (5–10 cm depth, hereafter “rhizosphere”). We obtained three biological replicates per compartment by placing soil into sterile tubes, either by scooping soil directly into the tube or using sterile plastic spoons. We preserved samples at -20°C in the field using an Engel MR040F portable freezer. At the end of each expedition, we transferred samples to an -80°C freezer in the laboratory.

Table 1 Summary of expedition coverage and environmental measurements.

Expedition	Season	Sites Visited	Environmental Measurements
Trip 1	Late Winter 2023	1–60, 4 special	Temp, Press, Hum.
Trip 2	Summer 2023	1–8	Temp, Press.
Trip 3	Winter 2024	1–60, 5 special	Temp, Press, Hum.
Trip 4	Summer 2024	1–60	Temp, Hum.
Trip 5	Autumn 2025	1–60, 2 special	Temp, Press, Hum.

We included positive and negative controls for each sampling campaign to mitigate and control for contamination artifacts. Positive controls comprised replicate tubes of Zymo Mock Community Standard (ZymoBIOMICS, cat #D6305). Negative controls consisted of sterilized soil treated with UV light for 4 hours.

2.2 DNA extraction and sequencing

We utilized 0.9–1.2 g of soil for DNA extraction using the PowerLyzer PowerSoil Kit (QIAGEN, catalog #12855-50) or the DNeasy PowerSoil Pro Kit (QIAGEN, catalog #47014). We homogenized all samples with the PowerLyzer 24 Homogenizer and completed extraction either manually or via the automated QIAcube Connect system. We quantified the eluted DNA using the Qubit HS dsDNA kit.

We amplified bacterial DNA using the Bakt_341F [18] and 806R [19] primer set to target the V3–V4 region of the 16S rRNA gene. We performed amplification using the Multiplex PCR kit (QIAGEN) and sequenced the resulting pools on an Illumina NovaSeq 6000 platform. We processed all samples in the PI’s laboratory at KAUST.

2.3 Sequencing data processing

We processed raw amplicon reads employing nf-core/Ampliseq (v2.14.0) [20, 21] and QIIME2 (v2024.10) [22]. We applied the DADA2 algorithm for error correction and denoising. We assigned taxonomy against the SILVA 138.2 reference database [23]. We retained amplicon sequence variants (ASVs) present in at least three samples with a total abundance greater than 50 reads. We identified and removed contaminant reads using the QIIME2 *decontam* module based on our control samples, applying a prevalence threshold of 0.52.

2.4 Taxonomy mapping and alignment

To unify taxonomic classifications derived from different databases, we developed an algorithm to map and align the NCBI Taxonomy (O_{ncbi}), the Genome Taxonomy Database (O_{gtdb}), and the iNaturalist taxonomy (O_{inat}). This alignment does not imply biological commitments but rather aims to integrate three different taxonomies for the purposes of the Rub’ al-Khali knowledge graph, specifically to enable queries within this project. The mapping process aligns the resolution of nomenclature divergence (e.g., lineage-specific alphanumeric suffixes in GTDB) and structural rank conflicts into a consolidated taxonomic knowledge graph O_{final} .

Let $O = (C, R, E)$ represent a taxonomy comprising a set of classes C , a set of taxonomic ranks R , and a set of subclass axioms E . We define a lexical normalization function $f_{norm} : C \rightarrow \Sigma^*$ that maps a class $c \in C$ to a canonical string by converting the primary label to lowercase, removing rank-specific prefixes (e.g., `d_`), and stripping alphanumeric lineage suffixes. Furthermore, let $rank : C \rightarrow R$ denote a function that retrieves the normalized taxonomic rank of a class, and let $parent : C \rightarrow C$ denote a function that retrieves the direct superclass of c .

The workflow initializes with a lexical alignment phase to generate a candidate seed set S_{lex} :

$$S_{lex} = \{(c_{src}, c_{tgt}) \in C_{ncbi} \times C_{gtdb} \mid f_{norm}(c_{src}) = f_{norm}(c_{tgt})\} \quad (1)$$

This initial lexical matching between NCBI and GTDB taxonomies produced 27,487 seed pairs. We subsequently partitioned S_{lex} through a structural verification step evaluating rank compatibility and topological support. A seed pair $(c_{src}, c_{tgt}) \in S_{lex}$ was classified as a strong anchor A_{strong} if $rank(c_{src}) = rank(c_{tgt})$ and $f_{norm}(parent(c_{src})) = f_{norm}(parent(c_{tgt}))$. Instances where ranks were compatible but parent labels diverged were classified as weak anchors A_{weak} ($|A_{weak}| = 4,347$), which signify potential nomenclature reclassifications.

We resolved the weak anchors utilizing a bottom-up inference strategy: we used the biological heuristic that identical species-level designations refer to the homologous entity regardless of divergent higher-rank classifications. For each $(c_{src}, c_{tgt}) \in A_{weak}$, we generated a structural hint $h = (parent(c_{src}), parent(c_{tgt}))$. We aggregated these hints and applied an equivalence axiom to higher-level taxa if the vote count for a structural hint exceeded our predefined threshold. This resolution step generated 1,514 parent hints.

Algorithm 1 details the formal procedure for merging the ontologies and subsequently grafting the iNaturalist taxonomy.

The alignment process successfully applied 27,487 verified anchors to bridge the NCBI and GTDB frameworks. Following the primary integration, we anchored the iNaturalist taxonomy (O_{inat}) into the merged taxonomy. We indexed the merged taxonomy (consisting of NCBI and GTDB taxonomies) utilizing the normalization function and iteratively matched iNaturalist classes. For singular matches, we directly appended equivalence axioms. When multiple candidate nodes shared a canonical label, we disambiguated the mapping using rank annotations. We explicitly copied all axioms from O_{inat} into O_{final} , effectively grafting unanchored nodes while retaining their subclass topology.

2.5 Environmental and climate data

We retrieved historical climate data from the Open-Meteo database [13] for each sampling site, yielding 12,936 monthly and 582 annual records encompassing temperature, total precipitation, rainfall, and relative humidity. Furthermore, field researchers manually classified each site according to its biome (e.g., desert biome) and specific environmental features (e.g., sand dune, saline pan, gravel, desert oasis). We subsequently mapped these classifications to the Environment Ontology (ENVO) [16].

Algorithm 1 Taxonomy Alignment and Integration Workflow

```
1: Input:  $O_{ncbi}$ ,  $O_{gtdb}$ ,  $O_{inat}$ 
2: Output:  $O_{final}$ 
3:  $S_{lex} \leftarrow \emptyset$ ,  $A_{strong} \leftarrow \emptyset$ ,  $A_{weak} \leftarrow \emptyset$ 
4: // Phase 1: Lexical Anchoring
5: for  $(c_{src}, c_{tgt}) \in C_{ncbi} \times C_{gtdb}$  do
6:   if  $f_{norm}(c_{src}) = f_{norm}(c_{tgt})$  then
7:      $S_{lex} \leftarrow S_{lex} \cup \{(c_{src}, c_{tgt})\}$ 
8:   end if
9: end for
10: // Phase 2: Structural Verification
11: for  $(c_{src}, c_{tgt}) \in S_{lex}$  do
12:   if  $f_{norm}(\text{parent}(c_{src})) = f_{norm}(\text{parent}(c_{tgt}))$  then
13:      $A_{strong} \leftarrow A_{strong} \cup \{(c_{src}, c_{tgt})\}$ 
14:   else
15:      $A_{weak} \leftarrow A_{weak} \cup \{(c_{src}, c_{tgt})\}$ 
16:   end if
17: end for
18: // Phase 3: Bottom-up Resolution
19:  $H \leftarrow \text{AggregateParentHints}(A_{weak})$ 
20:  $O_{merged} \leftarrow O_{ncbi} \cup O_{gtdb}$ 
21: for  $(c_{src}, c_{tgt}) \in A_{strong} \cup H$  do
22:    $O_{merged} \leftarrow O_{merged} \cup \{\text{EquivalentClasses}(c_{src}, c_{tgt})\}$ 
23: end for
24:  $O_{merged} \leftarrow \text{DisambiguateHomonyms}(O_{merged})$ 
25: // Phase 4: iNaturalist Grafting
26:  $O_{final} \leftarrow O_{merged} \cup O_{inat}$ 
27: for  $c_{inat} \in C_{inat}$  do
28:    $M \leftarrow \{c \in C_{merged} \mid f_{norm}(c) = f_{norm}(c_{inat})\}$ 
29:   if  $|M| = 1$  then
30:      $O_{final} \leftarrow O_{final} \cup \{\text{EquivalentClasses}(c_{inat}, m \in M)\}$ 
31:   else if  $|M| > 1$  then
32:      $m^* \leftarrow \text{DisambiguateByRank}(M, \text{rank}(c_{inat}))$ 
33:      $O_{final} \leftarrow O_{final} \cup \{\text{EquivalentClasses}(c_{inat}, m^*)\}$ 
34:   end if
35: end for
36: return  $O_{final}$ 
```

2.6 Schema validation

To ensure structural integrity within the RDF graph, we used Shape Expressions (ShEx) [24]. ShEx provides a formal structural schema language that validates the presence, cardinality, and datatype of RDF properties. We defined specific ShEx shapes for all core entity types in the knowledge graph. These entity types include sampling

sites, soil samples, DNA extracts, sequencing datasets, measurements, quality control records, and taxonomy mappings.

We integrated this schema validation into our deployment workflow using a Shape Map. The Shape Map functions as a binding directive, explicitly linking node selectors (e.g., all instances of the sampling site class) to their respective ShEx shape definitions. The workflow tests each newly generated segment of instance data against the schema. It terminates the deployment if it detects missing mandatory properties (e.g., GPS coordinates for a sampling site) or invalid datatypes. This constraint application guarantees that all generated data conforms to the expected structural properties. We implemented this validation using the Apache Jena ShEx library (version 4.10.0) invoked from custom Groovy scripts.

2.7 Logical consistency (OWL reasoning)

We verified the logical consistency of the entire knowledge graph using the ELK reasoner [25]. The ELK reasoner operates on the OWL 2 EL profile and detects unsatisfiable classes and contradictory axioms within ontologies. ELK implements a polynomial-time algorithm, facilitating the consistency validation of large knowledge bases. However, this performance benefit necessitates a tradeoff in logical expressivity, as the OWL 2 EL profile does not support complex constraints such as universal quantification or inverse object properties found in ontologies like the SemanticScience Integrated Ontology (SIO). We checked the unified knowledge graph to ensure that no individual belonged to disjoint classes and that property constraints remained logically sound. We employed ELK instead of more expressive reasoners (e.g., HermiT [26]) because verifying the large knowledge graph we created with a more expressive reasoners proved impossible due to computational complexity. This approach enabled rapid continuous integration during ontology development.

3 Knowledge representation

We developed a formal knowledge base to integrate the different types of data generated in this project. The semantic layer encodes the full provenance chain from physical sampling to digital data products, enabling cross-domain queries and ensuring data interoperability through shared vocabularies.

3.1 Architecture

The knowledge base is constructed in Description Logic [27] using a modular architecture that strictly separates the terminology (Terminological Box, TBox, or ontology) from the observed facts (Assertion Box, ABox, or domain knowledge). The TBox defines the ontology of the knowledge graph using OWL 2 DL [11], while the ABox contains the instance data generated from the expedition. We deployed the complete knowledge base across three named RDF graphs: an ontology graph (17,025 triples) containing the TBox, a knowledge base graph containing the asserted instance data, and a taxonomy reference graph containing the aligned taxonomy module.

3.2 Prefixes used in listings

For brevity, the Turtle and SPARQL listings throughout the paper use the prefix declarations of Table 2; these declarations should be prepended to any listing that the reader wishes to execute against the public endpoint. With these declarations in place, identifiers such as `rak:P100001` are abbreviated to `rak:P100001` and `sio:SIO_000291` to `sio:000291` without loss of information.

Table 2 Prefix declarations used in all Turtle and SPARQL listings.

Prefix	Namespace
<code>rak:</code>	<code>https://rubalkhali.science/kb/RAK_</code>
<code>sio:</code>	<code>http://semanticscience.org/resource/SIO_</code>
<code>pato:</code>	<code>http://purl.obolibrary.org/obo/PATO_</code>
<code>uo:</code>	<code>http://purl.obolibrary.org/obo/UO_</code>
<code>envo:</code>	<code>http://purl.obolibrary.org/obo/ENVO_</code>
<code>ncbitaxon:</code>	<code>http://purl.obolibrary.org/obo/NCBITaxon_</code>
<code>obo:</code>	<code>http://purl.obolibrary.org/obo/</code>
<code>rdf:</code>	<code>http://www.w3.org/1999/02/22-rdf-syntax-ns#</code>
<code>rdfs:</code>	<code>http://www.w3.org/2000/01/rdf-schema#</code>
<code>xsd:</code>	<code>http://www.w3.org/2001/XMLSchema#</code>
<code>geo:</code>	<code>http://www.opengis.net/ont/geosparql#</code>
<code>dcterms:</code>	<code>http://purl.org/dc/terms/</code>

3.3 Ontology engineering and design patterns

We use the Semanticscience Integrated Ontology (SIO) [14] as the upper-level framework. SIO provides general-purpose classes and relations applicable across scientific domains, ensuring semantic interoperability with other SIO-based datasets. We defined all domain-specific classes as subclasses of SIO classes, and all introduced relations are sub-relations of SIO relations. We mapped classes to SIO and maintained full semantic mappings in our public repository (see Section 7).

The core ontology comprises 302 distinct domain-specific classes integrated directly under the SIO framework. To accurately represent the complex relationships between environmental samples, laboratory processes, and bioinformatic workflows, we incorporated 4 object properties and 18 datatype properties. In total, the terminology model encapsulates 1,290 formal axioms. This compact semantic structure maintains precise alignment with our standardized schemas while minimizing redundant inferences.

3.3.1 Measurement pattern

We model a quantitative measurement using SIO’s ontology design pattern for measurement processes: a measurement is a process that generates a value for a specific quality of a target entity.

$$\text{MeasurementProcess} \sqsubseteq \text{Process} \quad (2)$$

$$\text{Quality} \sqsubseteq \text{Attribute} \quad (3)$$

$$\text{Quantity} \sqsubseteq \text{InformationContentEntity} \quad (4)$$

$$\text{MeasurementProcess} \sqsubseteq \exists \text{hasTarget.Entity} \sqcap \exists \text{hasOutput.Quantity} \quad (5)$$

$$\text{Quantity} \sqsubseteq \exists \text{hasValue.xsd:double} \sqcap \exists \text{hasUnit.Unit} \quad (6)$$

We identified all units with classes from the Units Ontology (UO) [28]. Qualities are expressed using the Phenotype and Trait Ontology (PATO) [29] and linked to SIO as subclasses of the *Quality* class. To illustrate this pattern, we provide an examples represented in Turtle format. The example (Listing 1) demonstrates an environmental temperature measurement taken at a sampling site.

Listing 1 Turtle representation of an environmental temperature measurement.

```
rak:P000001 a rak:0000006 ; # Measurement Process
  sio:000291 rak:1000001 ; # hasTarget
  sio:000229 rak:4000001 . # hasOutput (Value)

rak:4000001 a rak:0000010 ; # Quantity
  sio:000221 uo:0000027 ; # hasUnit (degree Celsius)
  rak:2000003 "20.7"^^xsd:double ; # hasValue
  sio:000215 rak:5000001 . # refersTo (Quality)

rak:5000001 a pato:0000146 ; # Temperature Quality
  sio:000011 rak:1000001 . # isAttributeOf
```

In words, an instance `rak:P000001` of a measurement process has as its target the sampling site `rak:1000001` and produces as output the quantity `rak:4000001`. That quantity carries the literal value 20.7, is reported in degrees Celsius (`uo:0000027`), and refers to a temperature quality `rak:5000001` which is in turn an attribute of the same site. Reading the chain in either direction therefore connects the numeric reading to both the measured site and the abstract quality that it quantifies.

3.3.2 Sampling pattern

A sampling event is a process that derives a sample from a feature of interest (the site).

$$\text{SamplingProcess} \sqsubseteq \text{Process} \quad (7)$$

$$\text{Sample} \sqsubseteq \text{MaterialEntity} \quad (8)$$

$$\text{SamplingProcess} \sqsubseteq \exists \text{hasTarget.Site} \sqcap \exists \text{hasOutput.Sample} \quad (9)$$

$$\text{Sample} \sqsubseteq \exists \text{isDerivedFrom.Site} \quad (10)$$

Listing 2 illustrates the semantic representation of a surface soil sample collected during Trip 1.

Listing 2 Turtle representation of a sampling process.

```

rak:P100001 a rak:0000024 ; # Sampling Process
  sio:000291 rak:1000001 ; # hasTarget
  sio:000229 rak:7000001 . # hasOutput (Collection)

rak:6000001 a rak:0000021 ; # Deep Soil Sample
  sio:000252 rak:1000001 . # isDerivedFrom

```

Read in prose, an instance `rak:P100001` of a sampling process targets the site `rak:1000001` and yields a collection `rak:7000001` as its output; the deep soil sample `rak:6000001` is in turn asserted to be derived from the same site. The redundancy between `hasTarget` on the process and `isDerivedFrom` on the sample is intentional: the former records the action, the latter records the resulting provenance edge directly on the material entity, so queries can traverse from sample to site without joining through the sampling event.

3.3.3 Experimental protocol pattern

We model experimental protocols as processes specified by a protocol entity. Each process (e.g., DNA extraction, library preparation) consumes inputs and produces outputs, explicitly linking agents and devices. The pattern instantiates SIO’s general *Process–hasInput/hasOutput–Object* idiom and adds a specification link to a reusable protocol description.

$$\text{ProtocolExecution} \sqsubseteq \text{Process} \quad (11)$$

$$\text{Protocol} \sqsubseteq \text{InformationContentEntity} \quad (12)$$

$$\text{ProtocolExecution} \sqsubseteq \exists \text{isSpecifiedBy}.\text{Protocol} \quad (13)$$

$$\text{ProtocolExecution} \sqsubseteq \exists \text{hasInput}.\text{MaterialEntity} \sqcap \exists \text{hasOutput}.\text{MaterialEntity} \quad (14)$$

Listing 3 demonstrates the formalization of a DNA extraction process.

Listing 3 Turtle representation of a DNA extraction protocol.

```

rak:P200001 a sio:000994 ; # Scientific Protocol Execution
  sio:000339 rak:DNA_ext_powersoil ; # isSpecifiedBy
  sio:000230 rak:6000001 ; # hasInput (Soil Sample)
  sio:000229 rak:6800001 . # hasOutput (DNA Extract)

rak:6800001 a rak:0000040 ; # DNA Extract
  sio:000008 rak:4900001 . # hasAttribute (Concentration Quality)

```

In words, the protocol-execution instance `rak:P200001` is specified by the named extraction protocol `rak:DNA_ext_powersoil`, takes the soil sample `rak:6000001` as input, and produces the DNA extract `rak:6800001` as output; the resulting extract carries a concentration quality (`rak:4900001`) which is itself quantified through a separate measurement process following the pattern of Equation 2.

3.3.4 Taxonomic abundance pattern

Taxonomic profiles represent collections of taxon observations containing associated relative and absolute abundances. Instead of modeling the feature table as a flat

matrix, we formalize the entire table as a bioinformatic workflow that produces a dataset containing individual taxonomic observations. Here, we use the merged taxonomy (see Section 2) to assign specific NCBI or GTDB classes as the target of each observation. Listing 4 shows how we map an entire sample’s feature table row into the knowledge graph.

For instance, the sequence analysis for a specific sample yields an absolute sequence count associated with a taxonomic class from the NCBI Taxonomy. We construct an abundance quality that targets the explicit taxonomic identifier, binding the observational count back to the biological workflow. Formally, the pattern instantiates SIO’s *Collection–hasMember–Thing* and *Quantity–refersTo–Quality* idioms:

$$\text{BioinformaticWorkflow} \sqsubseteq \text{Process} \quad (15)$$

$$\text{TaxonomicDataset} \sqsubseteq \text{Collection} \quad (16)$$

$$\text{BioinformaticWorkflow} \sqsubseteq \exists \text{hasInput.SequenceDataset} \sqcap \exists \text{hasOutput.TaxonomicDataset} \quad (17)$$

$$\text{TaxonomicDataset} \sqsubseteq \forall \text{hasMember.AbundanceObservation} \quad (18)$$

$$\text{AbundanceObservation} \sqsubseteq \text{Quantity} \sqcap \exists \text{refersTo.AbundanceQuality} \quad (19)$$

$$\text{AbundanceQuality} \sqsubseteq \exists \text{isAttributeOf.Taxon} \quad (20)$$

Listing 4 Turtle representation of taxonomic abundance formalized from a feature table.

```
rak:P290001 a rak:0000071 ; # Bioinformatic Workflow
  sio:000230 rak:7900001 ; # hasInput (FASTQ Dataset)
  sio:000229 rak:7290001 . # hasOutput (Taxonomic Dataset)

rak:7290001 a rak:0000074 ; # Taxonomic Dataset
  sio:000059 rak:4400001 . # hasMember (Observation Value)

rak:4400001 a sio:000070 ; # Quantity (Absolute Count)
  sio:000215 rak:5400001 ; # refersTo
  rak:2000021 "15"^^xsd:integer . # absolute read count

rak:5400001 a rak:0000078 ; # Absolute Abundance Quality
  sio:000011 ncbitaxon:286 . # isAttributeOf (Pseudomonas)
```

Verbalized, the bioinformatic-workflow instance **rak:P290001** consumes a FASTQ dataset (**rak:7900001**) and produces the taxonomic dataset **rak:7290001**; that dataset has as one of its members the quantity **rak:4400001**, which records an absolute read count of 15 and refers to an absolute abundance quality that is in turn an attribute of *Pseudomonas* (**obo:NCBITaxon.286**). One row of a feature table is therefore expanded into a four-node sub-graph that preserves both the classifier output and its provenance.

3.3.5 XRF geochemistry pattern

We model XRF analysis as a process that targets a soil sample and produces multiple measurement values, each corresponding to the concentration of a specific chemical analyte. We map each analyte to the Chemical Entities of Biological Interest (ChEBI)

ontology [15] as well as to the PubChem database [30]. Out of 93 monitored geochemical parameters, we mapped 80 analytes directly to corresponding ChEBI classes and aligned 92 parameters to PubChem Compound Identifiers. The XRF pattern is a specialization of the measurement pattern (Equation 2), with the target restricted to a soil sample and the produced quantity bound to an analyte-specific quality:

$$\text{XRFMeasurementProcess} \sqsubseteq \text{MeasurementProcess} \quad (21)$$

$$\text{XRFMeasurementProcess} \sqsubseteq \exists \text{hasTarget}.\text{SoilSample} \sqcap \exists \text{hasOutput}.\text{ConcentrationQuantity} \quad (22)$$

$$\text{ConcentrationQuantity} \sqsubseteq \text{Quantity} \sqcap \exists \text{refersTo}.\text{AnalyteQuality} \quad (23)$$

$$\text{AnalyteQuality} \sqsubseteq \exists \text{isAttributeOf}.\text{SoilSample} \quad (24)$$

Listing 5 illustrates an XRF measurement for a specific analyte.

Listing 5 Turtle representation of an XRF geochemistry measurement.

```
rak:P300001 a rak:0000080 ; # XRF Measuring Process
  sio:000291 rak:6000001 ; # hasTarget
  sio:000229 rak:4300001 . # hasOutput

rak:4300001 a sio:000070 ; # Quantity
  sio:000215 rak:5300001 ; # refersTo
  sio:000221 uo:0000187 ; # Percent unit
  rak:2000012 "58.2"^^xsd:double . # Concentration

rak:5300001 a rak:0000167 ; # Analyte Quality Class
  sio:000011 rak:6000001 . # isAttributeOf
```

In words, the XRF measurement process `rak:P300001` targets the soil sample `rak:6000001` and produces the quantity `rak:4300001`, which records a concentration of 58.2 percent (`uo:UO_0000187`) and refers to an analyte-specific quality `rak:5300001`. Because the analyte quality is typed with a dedicated subclass per element, the same pattern instance can be reused unchanged across all 91 geochemical analytes by varying only the quality class IRI.

3.3.6 Sampling site and biome assignment pattern

Each physical sampling location is represented as a persistent individual that is grounded in the Environment Ontology (ENVO) by linking to a biome and to one or more environmental features. We use both an existential and a universal restriction so that the asserted biome both holds and is the only admissible value, which preserves the closed-world intent of the field record.

$$\text{SamplingSite} \sqsubseteq \text{Site} \quad (25)$$

$$\text{SamplingSite} \sqsubseteq \exists \text{hasBiome}.\text{ENVO:Biome} \sqcap \forall \text{hasBiome}.\text{ENVO:Biome} \quad (26)$$

$$\text{SamplingSite} \sqsubseteq \exists \text{hasEnvironmentalFeature}.\text{ENVO:EnvironmentalFeature} \quad (27)$$

Listing 6 Turtle representation of a sampling site grounded in ENVO.

```
rak:1000001 a rak:0000002 ; # Sampling Site
  rdfs:label "Site 11" ;
  geo:asWKT "POINT(50.123 22.456)"^^geo:wktLiteral ;
  rak:2000050 envs:01000219 ; # hasBiome (desert biome)
  rak:2000051 envs:00000086 . # hasEnvironmentalFeature (sand dune)
```

In words, the sampling site `rak:1000001` carries a label, a WKT geometry, and is asserted both to belong to the ENVO desert biome and to exhibit a sand-dune environmental feature. The combined existential and universal restriction on `hasBiome` (Equation 25) ensures that any reasoner treats the asserted biome as the only admissible value for that site.

3.3.7 Sequencing workflow pattern

Sequencing data are produced by a chain of laboratory and computational processes that the knowledge graph represents as a sequence of *Process-hasInput/hasOutput-Object* steps: a DNA extraction process consumes a soil sample and produces a DNA extract; a library preparation process consumes the DNA extract and produces an amplicon library; a sequencing process consumes the amplicon library and produces a FASTQ dataset whose members are sequence reads. Each process is specified by a protocol and may carry agents and device participants, reusing the experimental protocol pattern of Equation 11.

$$\text{DNAExtractionProcess} \sqsubseteq \text{ProtocolExecution} \quad (28)$$

$$\text{DNAExtractionProcess} \sqsubseteq \exists \text{hasInput.SoilSample} \sqcap \exists \text{hasOutput.DNAExtract} \quad (29)$$

$$\text{LibraryPreparationProcess} \sqsubseteq \text{ProtocolExecution} \quad (30)$$

$$\text{LibraryPreparationProcess} \sqsubseteq \exists \text{hasInput.DNAExtract} \sqcap \exists \text{hasOutput.AmpliconLibrary} \quad (31)$$

$$\text{SequencingProcess} \sqsubseteq \text{ProtocolExecution} \quad (32)$$

$$\text{SequencingProcess} \sqsubseteq \exists \text{hasInput.AmpliconLibrary} \sqcap \exists \text{hasOutput.FASTQDataset} \quad (33)$$

$$\text{FASTQDataset} \sqsubseteq \text{Dataset} \sqcap \forall \text{hasMember.SequenceRead} \quad (34)$$

Listing 7 Turtle representation of the sequencing workflow from soil sample to FASTQ dataset.

```
rak:P200001 a rak:0000044 ; # DNA Extraction Process
  sio:000339 rak:L000001 ; # isSpecifiedBy (PowerSoil protocol)
  sio:000230 rak:6000001 ; # hasInput (Soil Sample)
  sio:000229 rak:6800001 . # hasOutput (DNA Extract)

rak:P250001 a rak:0000045 ; # Library Preparation Process
  sio:000339 rak:L000010 ; # Illumina 16S Prep Guide
  sio:000230 rak:6800001 ; # hasInput (DNA Extract)
  sio:000229 rak:6900001 . # hasOutput (Amplicon Library)
```

```

rak:P280001 a rak:0000046 ; # Sequencing Process
  sio:000230 rak:6900001 ; # hasInput (Amplicon Library)
  sio:000229 rak:7900001 . # hasOutput (FASTQ Dataset)

rak:7900001 a rak:0000064 ; # FASTQ Dataset
  rdfs:seeAlso <https://www.ebi.ac.uk/ena/browser/view/ERR1234567> .

```

Read as a chain, the DNA extraction process `rak:P200001` consumes the soil sample `rak:6000001` and produces the DNA extract `rak:6800001`; that extract is the input of the library preparation `rak:P250001`, which yields the amplicon library `rak:6900001`; finally the sequencing process `rak:P280001` consumes the library and produces the FASTQ dataset `rak:7900001`, which is cross-referenced to its ENA accession via `rdfs:seeAlso`. Each link in the chain is the same `hasInput/hasOutput` pattern, so a single SPARQL traversal can recover the full provenance from any node.

3.3.8 Sequencing quality control pattern

Sequencing quality control is represented as an additional computational process that takes a FASTQ dataset as input and produces multiple measurement values, each instantiating the generic measurement pattern of Equation 2. Metrics (sequence count, GC content, duplicate rate, read length) are split into forward (R1) and reverse (R2) subclasses so that the provenance of paired-end runs is preserved.

$$\text{SequencingQCProcess} \sqsubseteq \text{Process} \quad (35)$$

$$\text{SequencingQCProcess} \sqsubseteq \exists \text{hasInput.FASTQDataset} \sqcap \exists \text{hasOutput.QCMeasurementValue} \quad (36)$$

$$\text{QCMeasurementValue} \sqsubseteq \text{Quantity} \sqcap \exists \text{refersTo.QCQuality} \quad (37)$$

$$\text{QCQuality} \sqsubseteq \exists \text{isAttributeOf.FASTQDataset} \quad (38)$$

Listing 8 Turtle representation of a sequencing QC measurement (forward-read GC content).

```

rak:P350001 a rak:0000047 ; # Sequencing QC Process
  sio:000230 rak:7900001 ; # hasInput (FASTQ Dataset)
  sio:000229 rak:4500001 . # hasOutput (QC Value)

rak:4500001 a rak:0000086 ; # GC Content Value
  sio:000215 rak:5500001 ; # refersTo
  sio:000221 uo:0000187 ; # Percent
  rak:2000073 "54.2"^^xsd:double . # has GC content

rak:5500001 a rak:0000074 ; # GC Content Quality (Forward)
  sio:000011 rak:7900001 . # isAttributeOf (FASTQ Dataset)

```

In words, the QC process `rak:P350001` takes the FASTQ dataset `rak:7900001` as input and produces the measurement value `rak:4500001`, which records a GC content of 54.2 percent and refers to a forward-read GC-content quality `rak:5500001` that is in turn an attribute of the same FASTQ dataset. Splitting the quality class into forward and reverse subclasses preserves the read direction, so that paired-end QC metrics can be queried independently per orientation.

3.4 OWL and RDF implementation

We generated the OWL/RDF implementation using custom Groovy scripts utilizing the OWL API 5 [31]. The knowledge base employs a modular design, separating the base ontology, sites, samples, measurements, XRF data, DNA extraction records, sequencing metadata, quality control metrics, and taxonomic abundances into individual ABox modules [32].

We utilize the OpenLink Virtuoso triple store [33] for data hosting and query answering. Virtuoso features an inference engine configured with an OWL-Horst [34] rule set. This configuration enables the on-the-fly materialization of inverse properties, transitive closures, and property chains defined within the SIO ontology framework.

Prior to OWL-Horst inference, the base knowledge graph comprised 37,043,525 asserted triples across three named graphs (knowledge base 36,978,835; taxonomy 42,406; ontology 16,643), with taxonomy assertions accounting for approximately 36.5 million of the knowledge base graph. During deployment, the workflow applies OWL-Horst (pD*) forward-chaining materialization in three rounds: (1) ground rules that pair a single asserted triple with a TBox axiom (`rdfs:subClassOf`, `rdfs:subPropertyOf`, `rdfs:domain`, `rdfs:range`, `owl:inverseOf`, `owl:SymmetricProperty`, `owl:equivalentClass`, `owl:equivalentProperty`); (2) re-application of `subPropertyOf` and `subClassOf` over the triples produced in round 1; and (3) one-hop transitive closure for all `owl:TransitiveProperty` instances. This process generates 80,907,421 materialized triples, resulting in a total of 117,950,946 triples (a $3.18\times$ expansion). The materialization completes in approximately 114 minutes on a single server. The materialized triples are written to a dedicated named graph (`<https://rubalkhali.science/inferred/>`) so that they can be queried without per-query backward-chaining cost. Logging is temporarily disabled (`log-enable(2,1)`) during materialization to avoid exceeding Virtuoso's per-transaction log size limit. The dominant inferred predicates are `sio:isRelatedTo` (22.6M), `rdf:type` (14.6M), `sio:hasAttribute` (11.3M), and `sio:isAttributeOf` (5.7M), reflecting the propagation of SIO's property hierarchy and inverse properties across the ABox.

3.5 Schema definition

We used Shape Expressions (ShEx) [24] to define structural schemas for the core entity types. While OWL provides open-world reasoning and inference capabilities, it cannot readily enforce data completeness constraints. We developed ShEx schemas to explicitly validate the graph's structure, ensuring data quality and compliance before loading it into the database.

Our ShEx implementation defines shapes for Sampling Sites, Soil Samples, Environmental Measurements, DNA Extracts, and Sequencing Datasets. These shapes guarantee that essential attributes are invariably present. For instance, Listing 9 demonstrates how the schema enforces that every Sampling Site is explicitly typed and labeled, while optionally capturing geographic coordinates and textual descriptions.

Listing 9 ShEx shape for Sampling Site validation.

```

<#SamplingSiteShape> {
  rdf:type [rak:0000002] ; # Must be of type Sampling Site
  rdf:type . * ;          # May have other inferred types
  rdfs:label . ;          # Must have a label
  dct:terms:description xsd:string ? ; # Optional description
  geo:asWKT . ? ;         # Optional geographic coordinates
}

```

In words, the shape `<#SamplingSiteShape>` requires that every conforming node carry an explicit type assertion to `rak:0000002` (Sampling Site) and an `rdfs:label`; it permits any number of additional inferred types and optionally allows a free-text description and a WKT geometry. Records that omit the mandatory type or label are rejected before they reach the triple store.

4 Data records

We deposited the raw sequencing data generated in this study in the European Nucleotide Archive (ENA) under the umbrella project accession PRJEB104209. Ampli-con sequencing data reside under project PRJEB106069.

We made the knowledge graph accessible via a public SPARQL endpoint at <https://rubalkhali.science/sparql> and through a web interface at <https://rubalkhali.science/>. We maintain all source code for data processing, ontology generation, and validation at <https://github.com/bio-ontology-research-group/empty-quarter>.

We distribute the knowledge graph data files as OWL/XML and Turtle (.ttl) formats, structured by data type:

- `rubalkhali.owl` — Base ontology (TBox).
- `rubalkhali_sites.owl` — 70 sampling sites with coordinates and biome annotations.
- `rubalkhali_samples.owl` — 2,535 soil and plant samples with compartment types.
- `rubalkhali_measurements.owl` — Environmental measurements (temperature, pressure, humidity) and climate records (12,936 monthly, 582 annual).
- `rubalkhali_xrf.owl` — 6,674 XRF measurement values across 91 analytes from 249 measuring processes.
- `rubalkhali_dna.owl` — 2,410 DNA extracts with concentration data.
- `rubalkhali_sra.owl` — 1,242 sequencing run records linked to ENA accessions.
- `rubalkhali_qc.owl` — Sequencing quality metrics (GC content, duplicate rate, sequence counts).
- `rubalkhali_taxonomy_abox.ttl` — 1,401,008 taxon abundance observations with relative and absolute counts.

The data and source code to generate the Rub' al-Khali knowledge graph is available at [.](#)

5 Technical validation

We validated the knowledge graph using four complementary approaches, each targeting a distinct aspect of data quality: competency question evaluation via SPARQL

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queries, structural validation using Shape Expressions (ShEx), logical consistency checking with the ELK reasoner, and usability assessment through a web portal.

5.1 Competency questions and SPARQL queries

To validate whether the full knowledge graph can be used to answer queries about the Rub' al-Khali, we defined competency questions expressed as SPARQL queries [35] with known expected answers. We automated this validation process to enable regression testing during ontology updates. The validated competency questions include retrieving XRF measurements, listing sampling sites with coordinates, extracting DNA extraction data, tracing provenance from soil samples to sequencing runs, and querying taxonomic composition. The competency questions are designed to test questions that are needed for operating the website, and to support different types of queries within our dataset.

We show several example SPARQL queries. Listing 10 retrieves XRF measurements for a specific site, requiring traversal of the measurement process, quality, and value entities.

Listing 10 SPARQL query to retrieve field XRF measurements for Site 10. The query traverses from sample to measurement process to individual analyte concentrations.

```
PREFIX rak: <https://rubalkhali.science/kb/RAK_>
PREFIX sio: <http://semanticscience.org/resource/SIO_>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
SELECT ?processLabel ?analyte ?concentration ?unitLabel
WHERE {
  ?sample a rak:0000021 ;
          rdfs:label ?sampleLabel .
  FILTER(Regex(?sampleLabel, "Site 10"))
  ?value sio:000232 ?process .
  ?process rdfs:label ?processLabel .
  FILTER(Regex(?processLabel, "Field", "i"))
  ?sample sio:000008 ?quality .
  ?quality sio:000215 ?value .
  ?value rak:2000012 ?concentration .
  ?value sio:000221 ?unit .
  ?unit rdfs:label ?unitLabel .
  ?quality a ?qualityClass .
  ?qualityClass rdfs:label ?analyte .
  FILTER(?qualityClass != rak:0000029)
}
ORDER BY ?processLabel ?analyte LIMIT 100
```

Read in prose, the query first selects every soil sample whose label matches Site 10, then walks back from each measurement value to its producing process and forwards from the sample through its quality and value to the recorded concentration and unit. The final filter excludes the Light Elements pseudo-analyte (rak:0000029) so that only individually quantified elements appear in the result, and the query returns one row per process and analyte combination.

Listing 11 retrieves taxonomic abundance profiles, joining across bioinformatic workflows, FASTQ datasets, and taxon observations.

Listing 11 SPARQL query to retrieve taxonomic composition and abundance for all samples.

```
PREFIX rak: <https://rubalkhali.science/kb/RAK_>
```

```

PREFIX sio: <http://semanticscience.org/resource/SIO_>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
SELECT DISTINCT ?run ?taxonLabel ?count ?relativeAbundance
WHERE {
  ?proc a rak:0000071 ;
        sio:000230 ?fastq ;
        sio:000229 ?dsAbs , ?dsRel .
  ?dsAbs a rak:0000074 .
  ?dsRel a rak:0000075 .
  ?fastq rdfs:label ?runLabel .
  BIND(REPLACE(?runLabel, "FASTQ dataset for ", "") AS ?run)
  ?dsAbs sio:000059 ?valAbs .
  ?valAbs rak:2000021 ?count .
  ?valAbs sio:000215 ?qualAbs .
  ?qualAbs sio:000011 ?taxon .
  ?taxon rdfs:label ?taxonLabel .
  ?qualRel sio:000011 ?taxon ;
           a rak:0000072 ;
           sio:000214 ?valRel .
  ?valRel rak:2000020 ?relativeAbundance .
  FILTER(?taxon != ?fastq)
}
ORDER BY ?run DESC(?count) LIMIT 100

```

In words, the query identifies every bioinformatic workflow that produces both an absolute and a relative taxonomic dataset for the same FASTQ input, then for each absolute observation it retrieves the read count, the associated taxon label, and the matching relative abundance from the parallel relative dataset. Results are grouped by sequencing run and ordered by descending count, so the most abundant taxa per run appear first.

We added an additional query (Listing 12) to retrieve specific taxa abundances alongside environmental constraints, demonstrating the ability to cross-reference biological and environmental properties.

Listing 12 SPARQL query to retrieve *Pseudomonas* relative abundance at sites with high temperature.

```

PREFIX rak: <https://rubalkhali.science/kb/RAK_>
PREFIX sio: <http://semanticscience.org/resource/SIO_>
PREFIX pato: <http://purl.obolibrary.org/obo/PATO_>
PREFIX ncbitaxon: <http://purl.obolibrary.org/obo/NCBITaxon_>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
SELECT ?siteLabel ?temp ?relAbundance
WHERE {
  ?site a rak:0000002 ;
        rdfs:label ?siteLabel .

  # Environmental Temperature
  ?tempProcess sio:000291 ?site ;
              sio:000229 ?tempVal .
  ?tempVal rak:2000012 ?temp ;
           sio:000215 ?tempQual .
  ?tempQual a pato:0000146 .
  FILTER(?temp > 35.0)

  # Derived Sample and Taxon Abundance
  ?sample sio:000252 ?site .
  ?ext sio:000230 ?sample ; a rak:0000041 ; sio:000229 ?dna .
  ?libPrep sio:000230 ?dna ; a rak:0000062 ; sio:000229 ?lib .
}

```

```

?seq sio:000230 ?lib ; a rak:0000064 ; sio:000229 ?fastq .

?bioinf a rak:0000071 ;
    sio:000230 ?fastq ;
    sio:000229 ?dsRel .
?dsRel a rak:0000075 ;
    sio:000059 ?valRel .
?valRel rak:2000020 ?relAbundance ;
    sio:000215 ?qualRel .
?qualRel sio:000011 ncbitaxon:286 . # Pseudomonas
}

```

In words, the query first selects every sampling site at which the recorded environmental temperature exceeds 35 °C, then for each such site it walks the full provenance chain — soil sample, DNA extract, library preparation, sequencing run, FASTQ dataset, bioinformatic workflow, relative abundance dataset — down to the relative abundance value of *Pseudomonas* (`ncbitaxon:286`). Each row of the result therefore reports a single hot site together with its corresponding *Pseudomonas* relative abundance, demonstrating a join that crosses the environmental, geochemical, and biological domains in a single SPARQL query.

5.2 ShEx schema validation

We validated the structural integrity of the knowledge graph using Shape Expressions (ShEx), as described in Section 3.5. We defined ShEx shapes for all core entity types: sampling sites, soil samples, environmental measurements, DNA extracts, sequencing datasets, quality control records, and taxonomy observations. We bound each shape to its corresponding class instances via a Shape Map.

We implemented the validation using the Apache Jena ShEx library (version 4.10.0) invoked from Groovy scripts. We integrated ShEx validation into the automated deployment workflow: each time the workflow regenerates an ABox module, the validation step tests instances of the relevant class against their ShEx shape before loading the data into Virtuoso. The workflow terminates with an error if any instance violates the shape constraints (e.g., a sampling site lacking GPS coordinates or a DNA extract missing a concentration value). This mechanism functions as a regression test, ensuring that modifications to the ETL scripts do not introduce structural inconsistencies.

We validated the following eight RDF modules against their respective ShEx schemas: `rubalkhali_sites.owl`, `rubalkhali_samples.owl`, `rubalkhali_measurements.owl`, `rubalkhali_dna.owl`, `rubalkhali_xrf.owl`, `rubalkhali_taxonomy_abox.ttl`, `rubalkhali_sra.owl`, and `rubalkhali_qc.owl`. All eight modules passed validation without constraint violations.

5.3 Logical consistency

We verified the logical consistency of the knowledge graph using the ELK reasoner, as described in Section 2. We loaded the complete set of OWL modules (base ontology, all ABox files, and the taxonomy reference graph) into a single OWL ontology and ran ELK to check for unsatisfiable classes and contradictory axioms. The reasoner

confirmed that no class was reduced to the empty set (i.e., no class became equivalent to `owl:Nothing`) and that no individual belonged to disjoint classes.

We performed this consistency check as part of continuous integration during ontology development. Whenever we modified the TBox or introduced new ABox modules, we re-ran the ELK reasoner to detect regressions. We chose ELK over more expressive reasoners such as HermiT [26] because the size of the knowledge graph (exceeding 37 million triples) made reasoning in the full OWL 2 DL profile computationally infeasible.

We implemented the ELK consistency check via the OWL API (version 4.5.26) with the ELK OWL API module (version 0.4.3) in a Groovy script. We ran the reasoner in two configurations: (1) on the asserted knowledge graph prior to materialization, and (2) on the union of the asserted knowledge graph and the OWL-Horst materialized graph described in Section 3.4. Table 3 reports the results. In both cases the reasoner confirmed that the ontology is consistent and reported zero unsatisfiable classes. The post-materialization run loaded 81.3 million axioms (of which 78.1 million originate from the materialized ABox), discovered 74 additional classes (7,202 → 7,276) from the materialized type assertions, and required 63 seconds for the consistency check on a server with 1 TB RAM (395 GB peak resident set). These results confirm that the OWL-Horst entailments introduced no disjointness violations despite adding 5.7 million new individual type assertions.

Table 3 ELK reasoning performance before and after OWL-Horst materialization. All runs used OWL API 4.5.26 and ELK 0.4.3 on a server with 1 TB RAM (Java heap 850 GB).

Metric	Before	After
Modules loaded	8	10
Total axioms	224,775	81,273,368
Logical axioms	124,106	17,693,762
Classes	7,202	7,276
Named individuals	30,680	5,712,603
isConsistent	0.84 s	63.4 s
Class hierarchy	0.40 s	0.73 s
Unsatisfiable classes	0	0
Load time	675 s	7,585 s
Peak RSS	203 GB	395 GB
Wall time	688 s	7,722 s

5.4 Compute-time metrics

Table 4 summarizes the computational cost of the key deployment steps. OWL-Horst forward-chaining materialization (three rounds; see Section 3.4) runs in approximately 114 minutes and expands the knowledge graph from 37.0 million asserted triples

to 117.9 million (3.18 $\times$). ELK consistency checking over the asserted TBox takes under 2 seconds; the post-materialization run that includes the full materialized ABox requires 63 seconds but confirms the same result (zero unsatisfiable classes). Competency question latency on the Virtuoso endpoint ranges from 5 ms for simple site look-ups (CQ2) to over 1 second for queries that traverse inverse properties across the entire DNA extraction workflow (CQ5).

Table 4 Compute-time metrics for key deployment and validation steps.

Step	Time
RDF module loading (Virtuoso, 9 ABox files)	140 s
OWL-Horst materialization (3 rounds)	114 min
ELK classification (asserted only)	1.6 s
ELK classification (with materialized ABox)	64 s
CQ1 – XRF measurements, Site 10	23 ms
CQ2 – Sampling sites + GPS	5 ms
CQ3 – Biomes per site	5 ms
CQ4 – Light Elements percentages	8 ms
CQ5 – DNA extraction records	1,342 ms
CQ6 – Provenance chain	255 ms
CQ7 – Taxonomic composition	>300 s

The taxonomy composition query (CQ7) exceeds a 5-minute timeout because it joins absolute and relative abundance datasets across 1.4 million observations. This query is not used by the web portal; the portal retrieves taxonomic data per sample rather than across all runs, which completes in sub-second time.

5.5 Usability validation (web portal)

We developed a web portal (<https://rubalkhali.science/>) to provide user-friendly access to the data. The application architecture provides an end-to-end validation of the knowledge graph: the backend dynamically generates SPARQL queries to fetch content directly from the knowledge graph in real time without maintaining a separate relational database. Every page rendered constitutes a functional test of the underlying data. This architecture ensures that the website always reflects the current state of the knowledge graph, and any data integrity problem manifests as a visible error on the portal.

5.6 FAIR compliance

The design decisions described in the preceding sections collectively ensure that the dataset satisfies the FAIR Data Principles [36]:

- **Findable:** We assigned globally unique IRIs under the <https://rubalkhali.science/kb/> namespace to all named individuals. We registered the dataset with ENA under accession PRJEB104209.
- **Accessible:** We made data available via a standard SPARQL 1.1 endpoint and a web interface. We deposited raw sequences in ENA.
- **Interoperable:** We employ established shared vocabularies (SIO, ENVO, ChEBI, PATO, UO, NCBI Taxonomy) and standard formats (OWL 2 DL, RDF/Turtle, SPARQL).
- **Reusable:** We licensed the data under CC-BY 4.0 and described it with rich provenance metadata encoding the complete chain from physical sample to digital data product. We open-sourced all generation code.

6 Usage notes

The knowledge graph is publicly accessible via the SPARQL endpoint at <https://rubalkhali.science/sparql>. Users can submit SPARQL 1.1 queries using standard clients (e.g., `curl`, Python’s `SPARQLWrapper`, R’s `SPARQL` package). The web portal at <https://rubalkhali.science/> provides a graphical interface for browsing sites, samples, and measurements without requiring SPARQL expertise.

The semantic structure of the knowledge graph enables cross-domain queries. For example, researchers investigating the relationship between soil geochemistry and microbial diversity can construct a single query that retrieves calcium concentrations alongside Shannon diversity indices computed from taxon abundances. Executing this query in a relational database would necessitate joining at least four separate tables.

The modular OWL architecture permits extending the knowledge graph with new data types, such as metabolomics or remote sensing data. Users can introduce new classes as subclasses of existing SIO entities and generate corresponding ABox modules. We automate the entire lifecycle—from data generation through validation to database loading—using the provided deployment workflow (`deploy_onto.sh`).

6.1 Design rationale

We adopted the Semanticscience Integrated Ontology (SIO) as the upper-level framework because it offers a small set of relations (**hasInput**, **hasOutput**, **hasTarget**, **hasMember**, **refersTo**, **hasValue**) that generalize across sampling, measurement, and provenance patterns. Sample collection, XRF analysis, DNA extraction, sequencing, and taxonomic classification are all instantiations of the same **Process--hasInput/hasOutput--Object** idiom; quantitative results are uniformly represented as **Quantity refersTo Quality** pairs with explicit units. This uniformity enables cross-domain SPARQL queries that traverse from a climate observation through a sampling site to a taxon abundance without per-domain join logic. Recent work on simplified upper-level ontologies, such as SULO [37], takes this argument further by strengthening fundamental relations and reducing reliance on specialized predicates; our patterns are compatible with this direction.

We chose ShEx for ingest validation because the ETL is batch-oriented and the validator runs in the same Groovy/Jena toolchain that produces the RDF. SHACL is

an obvious alternative supported in-database by several triple stores; a future port of the shapes from ShEx to SHACL would enable in-database validation at load time. We deployed Virtuoso as the triple store because it is available as free software under the GNU General Public License, supports OWL-Horst forward-chaining materialization, and has demonstrated operational maturity in large-scale linked-data deployments.

Full OWL 2 DL reasoning over the combined TBox and the ABox is not computationally feasible with HermiT on commodity hardware. We therefore use ELK [25] to validate the logical consistency of the knowledge graph (confirming zero unsatisfiable classes both before and after materialization). For query answering, we rely on OWL-Horst forward-chaining materialization in Virtuoso, which physically expands the graph from 37 million asserted triples to approximately 118 million triples by applying subclass, subproperty, inverse, symmetric, equivalence, and transitivity rules. Two decades of work on scaling OWL reasoning—materialization-based approaches [34], query rewriting, modular reasoning, and consequence-based calculi such as ELK itself—provide several alternative points in the design space, and a more expressive reasoner could be applied to a sub-module of the TBox in future work.

6.2 Limitations and outstanding challenges

Several challenges remain open. First, more complex reasoning—for example over disjunctions arising from ENVO biome classifications or negative axioms about taxon absence—would require either a different reasoner or the extraction of a sub-module amenable to a more expressive calculus. Second, the resource depends on upstream ontologies (SIO, ChEBI, ENVO, NCBI Taxonomy) that evolve independently; keeping the deployed graph aligned with their releases is an ongoing maintenance task and a known reproducibility risk for any SPARQL query that bottoms out in a specific external IRI.

6.3 Sustainability and updates

The dataset is versioned and archived on Zenodo, and the deployment workflow is reproducible from the public repository. We intend to extend the resource as additional field expeditions are processed: new trips will be added as additional ABox modules without changes to the upper-level patterns. Upstream ontology updates will be tracked at scheduled intervals; re-materialization is scripted and runs as part of the deployment workflow, so an upstream release can be absorbed by re-running the workflow. Where an upstream change would alter the semantics of an existing query, we will record the affected competency questions in the changelog so that downstream users can audit the impact.

7 Code availability

All source code for data processing, ontology engineering, RDF generation, validation, and the web portal is openly available at <https://github.com/bio-ontology-research-group/empty-quarter>. The repository includes Groovy scripts

utilizing the OWL API 5 for OWL/RDF generation, Python scripts for data processing and statistical analysis, ShEx shape definitions for structural validation, SPARQL competency queries, React/FastAPI web application source code, and Docker Compose configurations for deployment.

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Author contributions

Add CRediT author contribution statements

Competing interests

The authors declare no competing interests.

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